Machine Learning on Customer Behaviour

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Understanding and predicting customer behaviour is critical across domains such as e-commerce, digital marketing, and consumer analytics. The goal of this project is to predict whether a user made a purchase based on session-level data, which includes behavioural signals like time spent on site, pages viewed, and ad engagement.

Approach:-

- Histograms, boxplots, and pairplots were used to visualize feature distributions. A correlation heatmap was used to identify relationships with the target variable.
- In the given data, two columns 'Referral' and 'Last_Ad_Seen' are non-numerical. To use that data in the algorithm I have converted them into numerical data using get_dummies() function.
- To perform machine learning I have chosen Logistic Regression and Random Forest algorithms. These both models are quite proven to be quite good for binary classifications.

1. For Logistic Regression,

Assumptions: Linearity between features and log-odds of the target

Pros: Fast, interpretable, good baseline

2. For Random Forest,

Assumptions: None strictly; robust to outliers and irrelevant features

Pros: Nonlinear modelling, feature importance, handles noise well

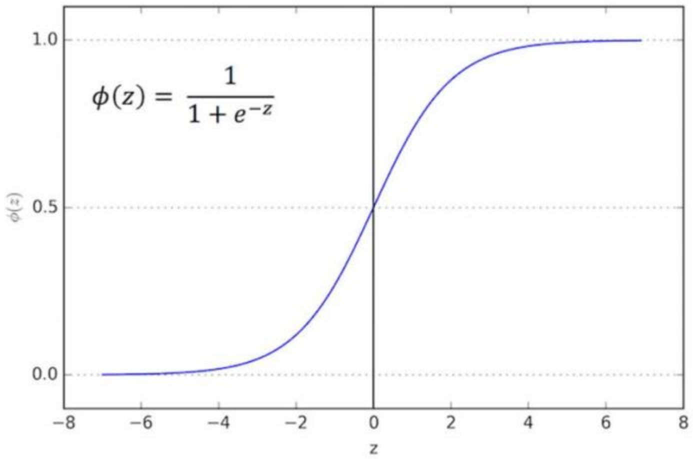
- Both models were implemented using scikit-learn.
- Grid Search was used to tune hyperparameters like regularization strength for Logistic Regression and tree depth and number of estimators for Random Forest. Optimal values improved performance metrics for both models. But it's quite a time taking process.

Brief Mathematics:-

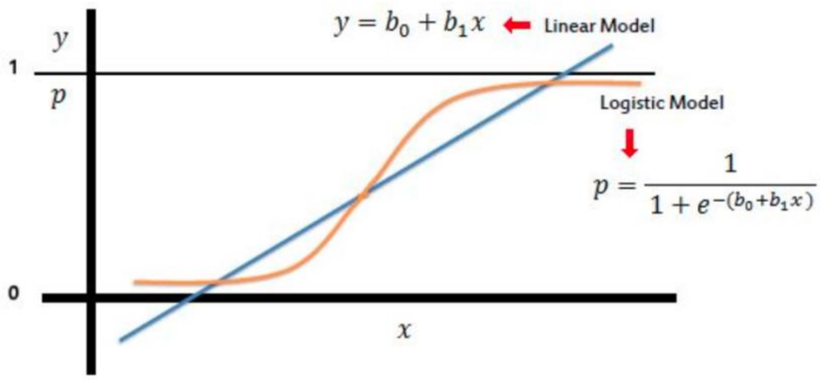
1. <u>Logistic Regression</u>

Logistic regression allows us to solve classification problems, where we are trying to predict discrete categories. The convention for binary classification is to have two classes 0 and 1. We can't use a normal linear regression model on binary groups. It won't lead to a good fit.

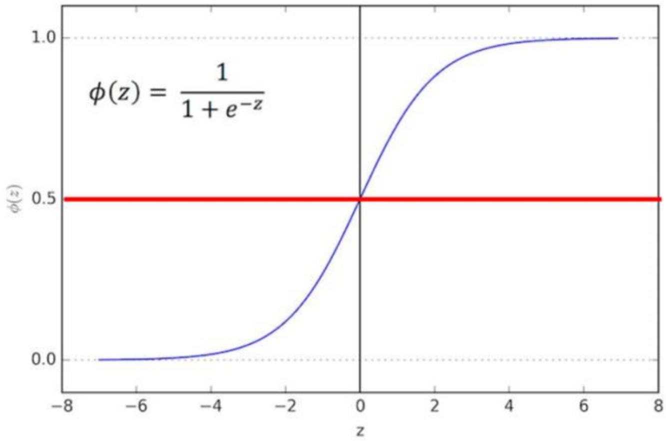
- The Sigmoid (aka Logistic) Function takes in any value and outputs it to be between 0 and 1.



- This means we can take our Linear Regression Solution and place it into the Sigmoid Function.



- This results in a probability from 0 to 1 of belonging in the 1 class.
- We can set a cut-off point at 0.5, anything below it results in class 0, anything above is class 1.
- We use the logistic function to output a value ranging from 0 to 1. Based off of this probability we assign a class.

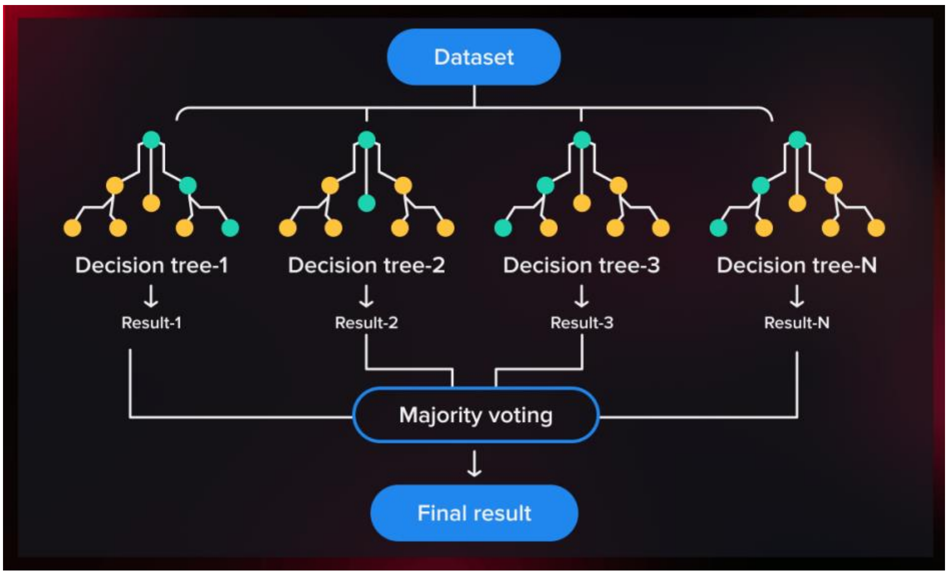


- Likelihood function

$$\ell(\beta_0, \beta_1) = \prod_{i:y_i=1} p(x_i) \prod_{i':y_{i'}=0} (1 - p(x_{i'})).$$

2. <u>Random Forest</u>

We can use many trees with a random sample of features chosen as the split.



- A new random sample of features is chosen for every single tree at every single split. Entropy and Information Gain are the Mathematical Methods of choosing the best split.

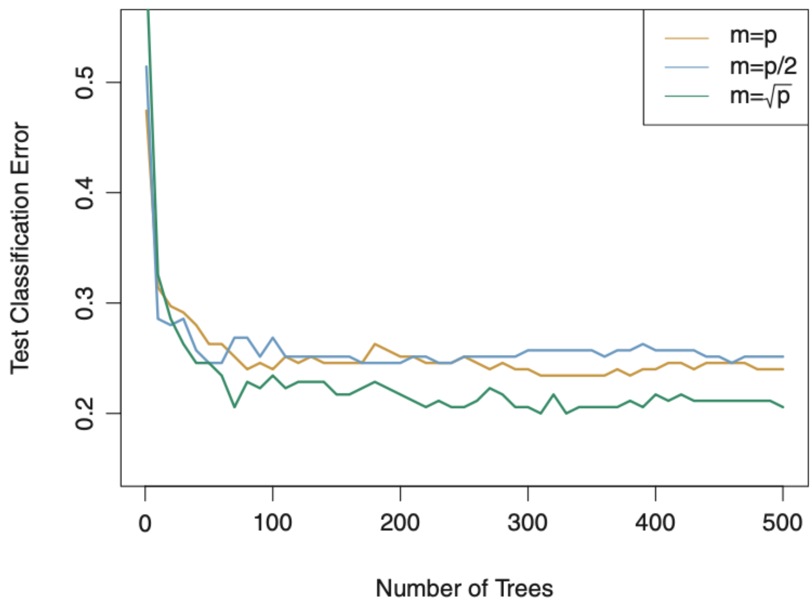
$$Entropy:$$

$$H(S) = -\sum_i p_i(S)\, log_2\, p_i(S)$$

$$Information\ Gain:$$

$$IG(S, A) = H(S) - \sum_{v \in Values(A)} \frac{|S_v|}{S} H(S_v)$$

- For classification, m is typically chosen to be the square root of p.



- Suppose there is one very strong feature in the data set. When using “bagged” trees, most of the trees will use that feature as the top split, resulting in an ensemble of similar trees that are highly correlated.
- Averaging highly correlated quantities does not significantly reduce variance.
- By randomly leaving out candidate features from each split, Random Forests "decorrelates" the trees, such that the averaging process can reduce the variance of the resulting model.

Performance Analysis:-

		predicted condition	
	total population	prediction positive	prediction negative
true condition	condition positive	True Positive (TP)	False Negative (FN) (type II error)
	condition negative	False Positive (FP) (Type I error)	True Negative (TN)

- Accuracy in classification problems is the number of correct predictions made by the model divided by the total number of predictions.
- Recall is the number of true positives divided by the number of true positives plus the number of false negatives.
- Precision is defined as the number of true positives divided by the number of true positives plus the number of false positives.

$$F_1 = 2 * \frac{precision * recall}{precision + recall}$$

1. Logistic Regression

```
Confusion matrix:
 [[279  35]
 [ 66  70]]
              precision    recall  f1-score   support

           0       0.81      0.89      0.85       314
           1       0.67      0.51      0.58       136

    accuracy                           0.78       450
   macro avg       0.74      0.70      0.71       450
weighted avg       0.77      0.78      0.77       450
```

Best hyperparameter found upon Grid Search: C=1 and max_iter=1000

2. Random Forest

```
Confusion matrix:
 [[272  42]
 [ 50  86]]
              precision    recall  f1-score   support

           0       0.84      0.87      0.86       314
           1       0.67      0.63      0.65       136

    accuracy                           0.80       450
   macro avg       0.76      0.75      0.75       450
weighted avg       0.79      0.80      0.79       450
```

Best hyperparameter found upon Grid Search: max_depth=10, min_samples_split=5 and n_estimators=200

Comparison:-

- Random Forest outperformed Logistic Regression in both accuracy and F1 score.
- Its ability to handle non-linear relationships and feature noise made it more robust.
- Logistic Regression is preferred when model interpretability and training speed are critical.
- Overall, Random Forest is more suitable for this problem due to dataset complexity.

Conclusion:-

This project demonstrated how to build a predictive model for customer purchase prediction using real-world data. A comprehensive EDA helped uncover structure in the data, and two ML models were compared in detail. Random Forest emerged as the better performing model, showing its strength in handling noisy, complex datasets.

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